

Ceph: A Storage and Backup Solution for Every Need

Backup and storage solutions assume paramount importance these days when data piles up in terabytes and petabytes, and the loss of it can be catastrophic. Ceph is the next generation, open source, distributed object store based low cost storage solution for petabyte-scale storage.



Storage is a key requirement these days and has to be thoroughly planned, keeping in mind performance, future enhancements, scalability and availability, as all these factors directly impact the functioning of the system. With the recent advances in cloud computing, the need for storage has been broadly classified into three major types—object store, block store and file system store. Ceph is an open source, reliable and easy-to-manage, next-generation distributed object store based storage platform that provides different interfaces for the storage of unstructured data from applications.

Ceph, being an object store, provides an interface for all three types of data storage, i.e., object store, block store and file based store, hence giving a unified storage platform. Ceph is the best example of software-defined storage (SDS). Analogous to the principles of cloud computing, Ceph scales out using commodity hardware, considering the challenges of scaling up. This helps us define different hardware for different workloads, and manage multiple nodes as a single

entity. With distributed storage, Ceph allows the distribution of the load, which delivers the best possible performance from low cost devices. All this contributes to low cost hardware, high data efficiency, broader storage use cases and greater performance. These are some of the major benefits of Ceph, because of which it has been widely adopted recently by enterprises and cloud service providers. Major storage vendors like SanDisk and other enterprises like Red Hat have contributed significantly to Ceph's development. So let us take a closer look at it.

Architecture and design

Ceph is a scalable storage cluster based on RADOS (reliable, autonomic distributed object store), which is a self-healing and self-managing distributed object storage system. It has two types of nodes, Ceph Monitor and Ceph OSD. A Ceph Monitor node maintains the state of the cluster map to check for node availability, and provides the cluster map to the cluster clients. A Ceph OSD Daemon is an intelligent peer,